CAMPUS ASSESSMENT - BUILD BRIEF

Price Alert Tool

Build a small command-line tool that reads daily prices for several instruments, applies alert rules from a configuration file, and prints which alerts fired on which day. You may use any programming language and any AI assistant; we expect you to. What we evaluate is how you verified the result and how well you understand what you submit.

Inputs

Prices file (CSV, one row per instrument per trading day, dates may be out of order):

```
date,instrument,close
2026-09-01,INFY,1520.50
2026-09-01,TCS,3890.00
2026-09-02,INFY,1534.10
2026-09-02,TCS,3902.25
2026-09-03,INFY,1498.00
```

Rules file (JSON, any number of rules):

```
[
  {"id": "r1", "instrument": "INFY",
   "type": "level", "direction": "below",
   "level": 1500},
  {"id": "r2", "instrument": "TCS",
   "type": "ma_cross",
   "short": 5, "long": 20,
   "direction": "above"}
]
```

A **level** rule fires on the first day the close crosses the level in the given direction (it was on the other side the previous day). An **ma_cross** rule fires on the day the short moving average crosses the long one in the given direction. Moving averages need a full window of days before they are defined.

Output

```
2026-09-03  r1  INFY  level below 1500
2026-09-24  r2  TCS   ma_cross 5/20 above
```

One line per alert, ordered by date then rule id. Print nothing for days on which no rule fires.

Rules of the exercise

- **Any language, any assistant.** Use whatever you are productive with. Working with an AI assistant is expected, not penalised.
- **It must run from a clean download with one command**, for example `python alerts.py prices.csv rules.json`. Say what to install, if anything.
- **Include tests.** We will read them before we read the code. Think about what could go wrong with real data.
- **Include a README** of at most one page: how to run it, how you used the assistant (the prompts in brief), what the assistant got wrong, and what you changed and why.
- **Submit by 14:00 tomorrow** as a zip file or a repository link to the Wissen coordinator. Late submissions are not evaluated.
- The work must be yours. Help from other people is not permitted; help from AI tools is.

How it is evaluated

- The tests you wrote and the cases they cover.
- Whether the tool runs correctly on a prices file we bring, which will contain conditions you have not seen.
- The honesty and precision of the README.
- Your presentation: six minutes on what it does, how you verified it, and one thing the assistant got wrong; then six minutes of questions in which we run the tool with you.

A NOTE

A polished tool you cannot explain scores below a rough one you understand completely. Volume of code earns nothing. Bring your own laptop to the presentation if you have one.